Cyber Security: Principles and Practices [3 Marks]

Question - 2

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Introduction to Cyber Security

- Cyber security is the practice of protecting systems, networks, and programs from digital attacks.
- Importance:
  - Protect sensitive information
  - Prevent financial and reputational loss
  - Ensure safe communication over networks
- Cyber threats include malware, phishing, ransomware, and data breaches.



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Comparison of Security Measures

Feature	Traditional IT	Cyber Security Approach
Access Control	Physical locks	Multi-factor authentication
Data Protection	Local backups	Encryption, Cloud backups
Threat Detection	Manual	Real-time monitoring
Incident Response	Reactive	Proactive strategies



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Blocks and Highlights Example

Some critical terms can be highlighted to emphasize importance.

Confidentiality

Ensures that sensitive information is accessed only by authorized individuals.

Integrity

Maintains accuracy and trustworthiness of data and prevents unauthorized modification.

Examples

Availability ensures systems and data are accessible when needed by authorized users.



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Cryptography Basics

Definition

Encryption is the process of converting plain text into ciphertext using an algorithm and a key.

Definition

Decryption is the process of converting ciphertext back into plain text using a key.



Hash Functions

Theorem

A cryptographic hash function produces a fixed-size output for any input and is irreversible.

Lemma

Two different inputs must not produce the same hash (collision resistance).



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Cryptography Equations

Equation 1: Symmetric Encryption

$$C = E_K(M)$$

Equation 2: Symmetric Decryption

$$M = D_K(C)$$

Equation 3: Hashing

$$H = h(M)$$



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Conclusion

- Cyber security is essential for protecting digital assets.



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- Combining preventive, detective, and corrective measures improves security posture.



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- Cryptography, algorithms, and network monitoring form the backbone of cyber security practices.



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- Cyber security is essential for protecting digital assets.
- Combining preventive, detective, and corrective measures improves security posture.
- Cryptography, algorithms, and network monitoring form the backbone of cyber security practices.
- Be aware of evolving threats to maintain strong defenses.



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References I

This presentation is based on the following references:

- [2] for basic LaTeX concepts.
- [1] for fundamental TeX understanding.



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Thank You!

Any Questions?



